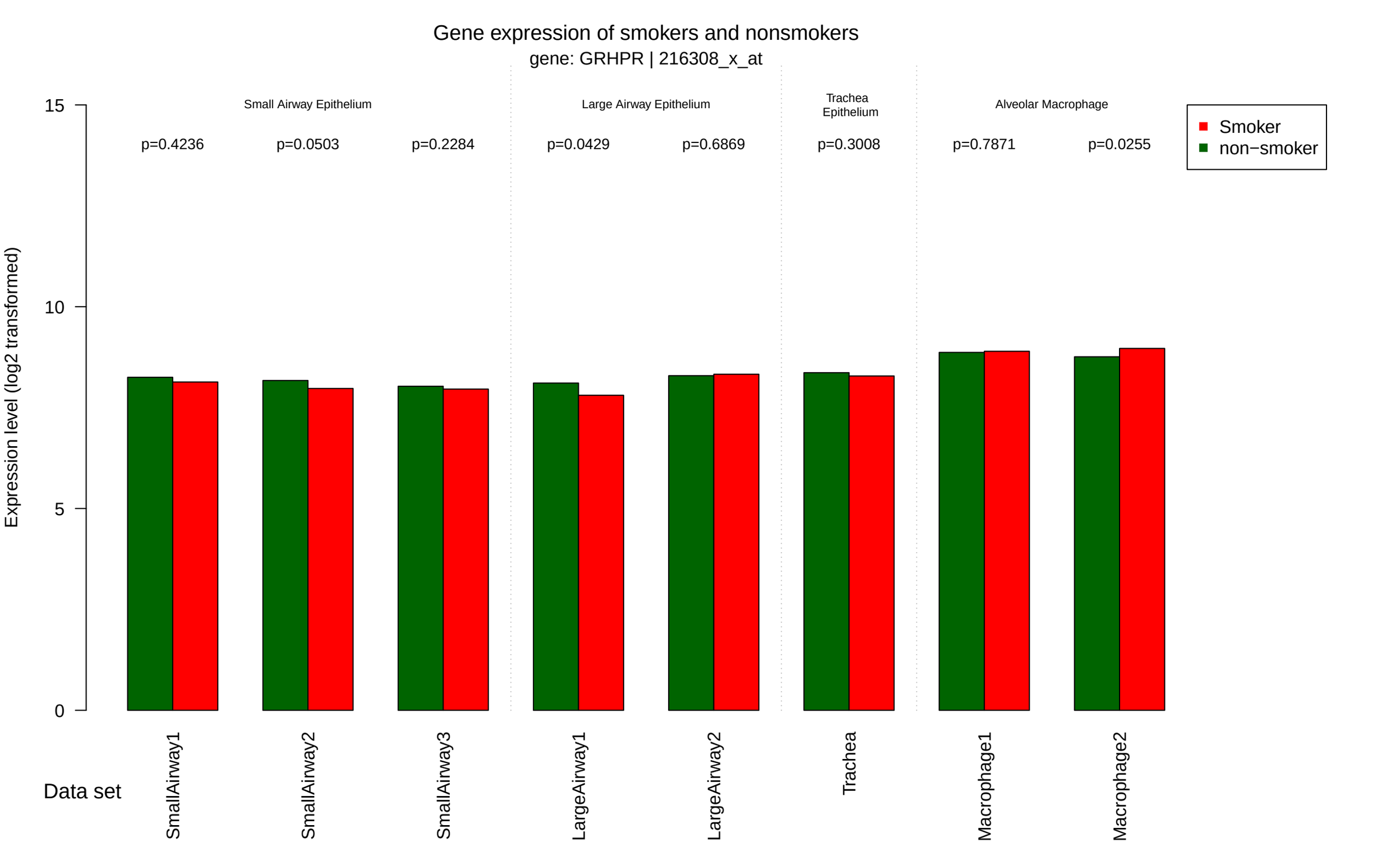
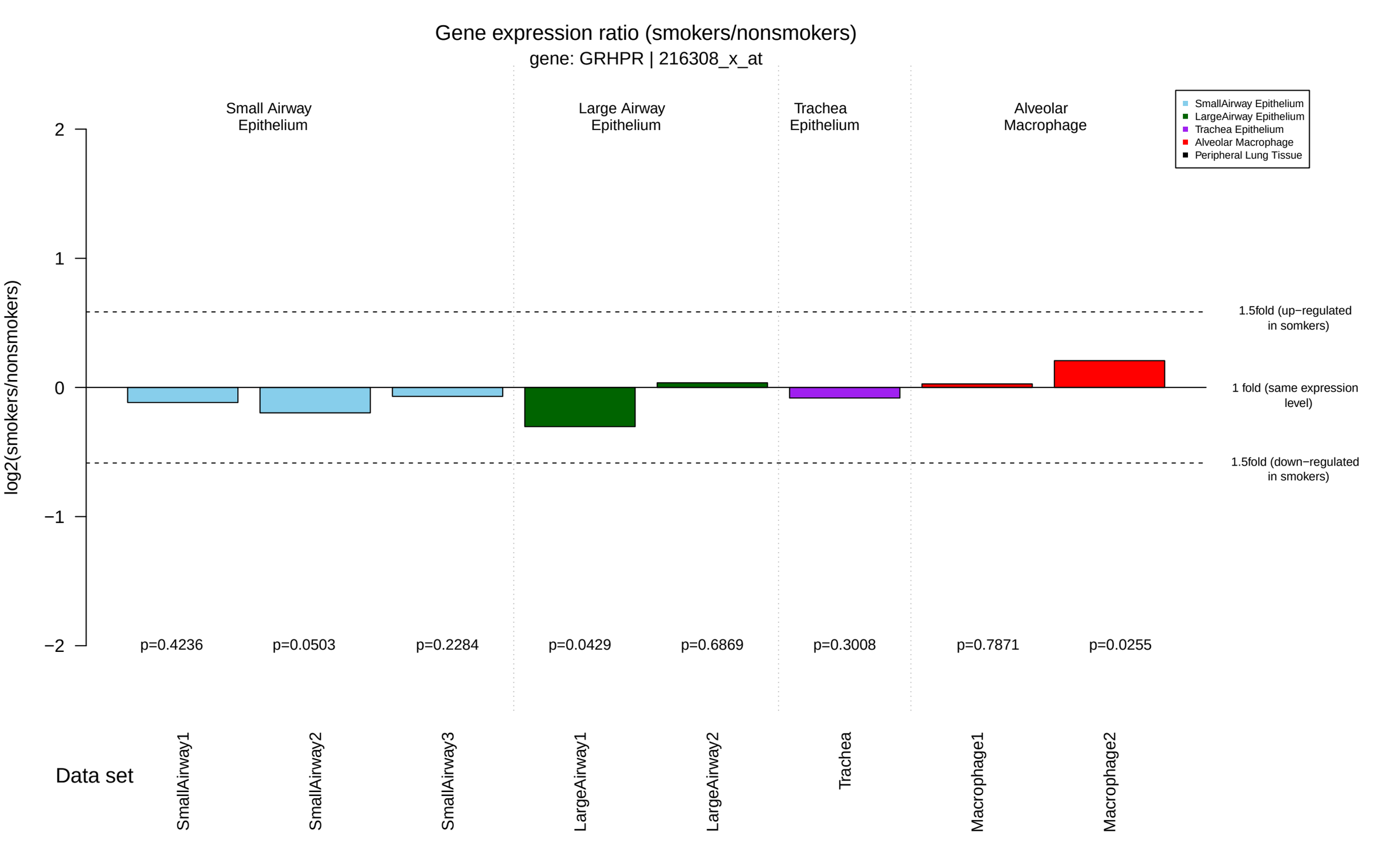
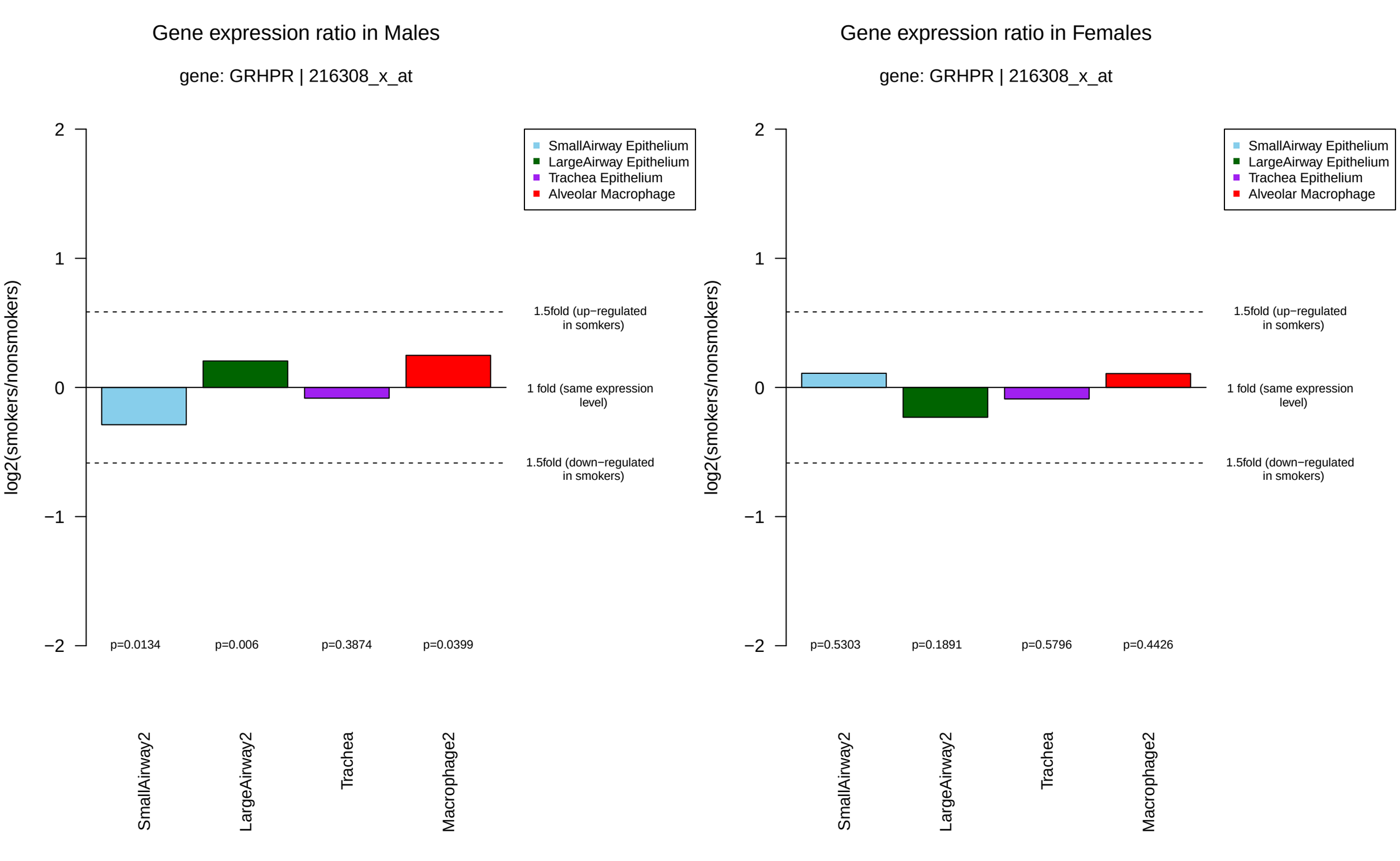
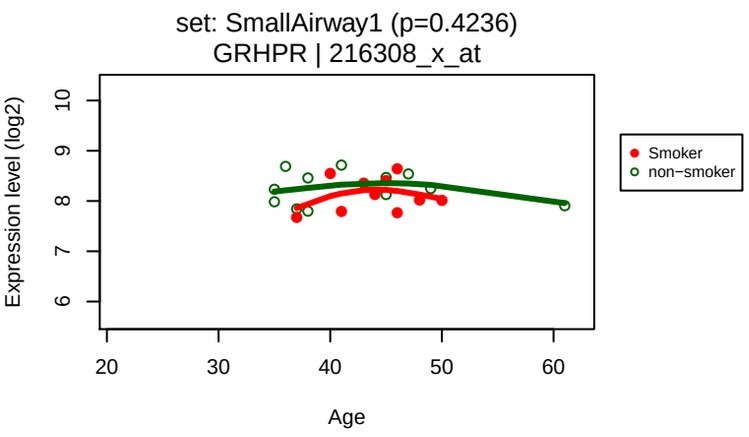
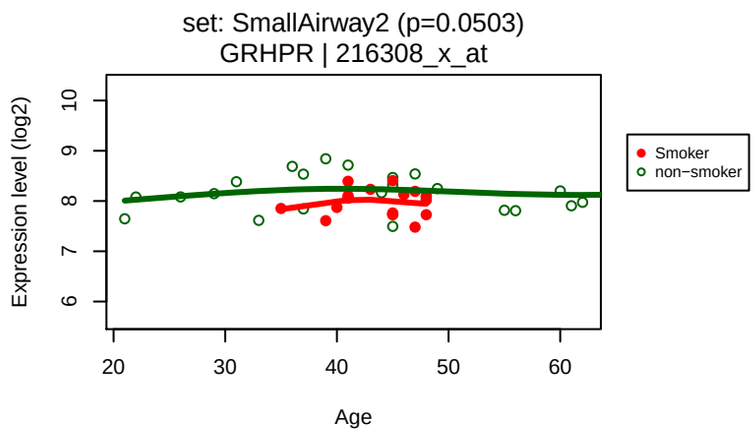
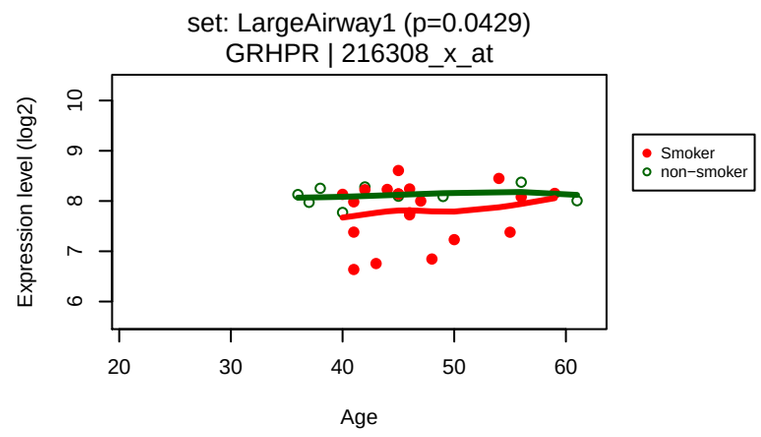

Gene expression of smokers and nonsmokers
gene: GRHPR | 216308_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4236
p=0.0503
p=0.2284
p=0.0429
p=0.6869
p=0.3008
p=0.7871
p=0.0255
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

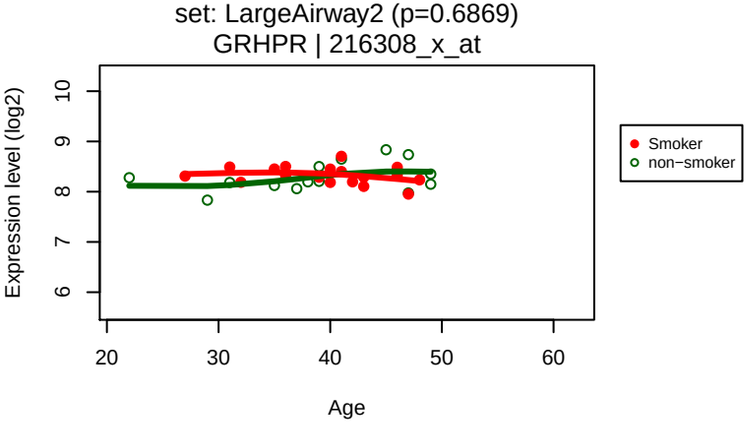

Gene expression ratio (smokers/nonsmokers)
gene: GRHPR | 216308_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4236
p=0.0503
p=0.2284
p=0.0429
p=0.6869
p=0.3008
p=0.7871
p=0.0255
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

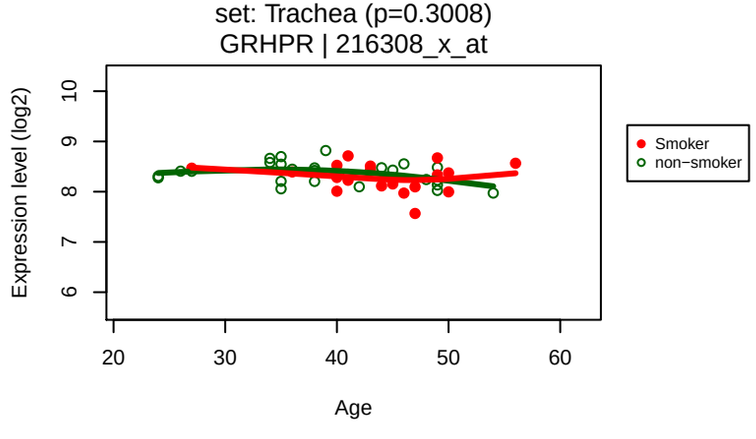

Gene expression ratio in Males
gene: GRHPR | 216308_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0134
p=0.006
p=0.3874
p=0.0399
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRHPR | 216308_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5303
p=0.1891
p=0.5796
p=0.4426
SmallAirway2
LargeAirway2
Trachea
Macrophage2

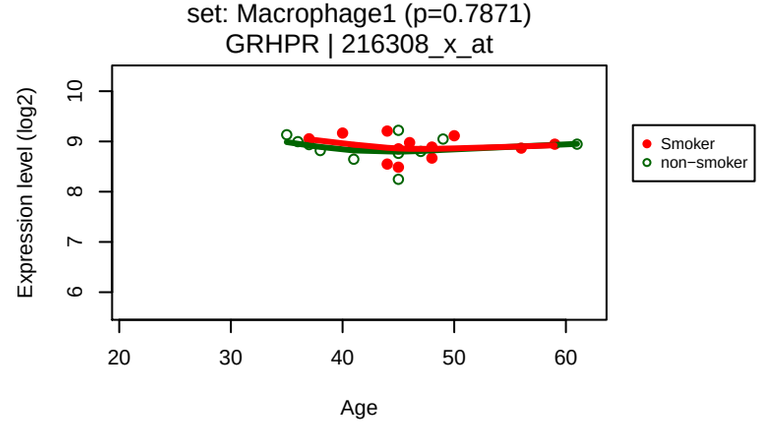

set: SmallAirway1 (p=0.4236)
GRHPR | 216308_x_at
Expression level (log2)
Age
Smoker
non−smoker

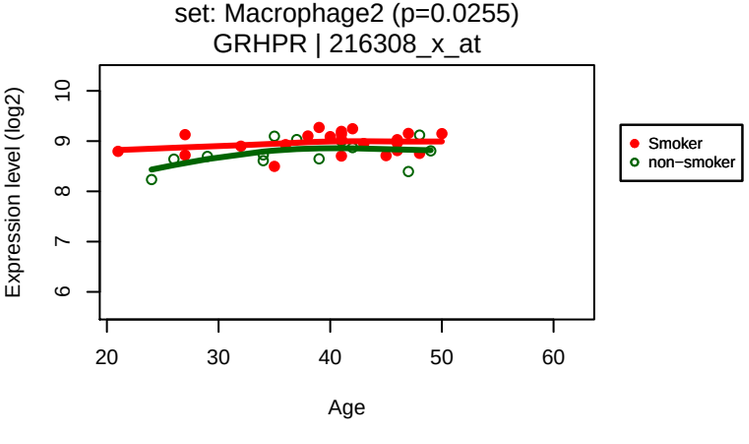

set: SmallAirway2 (p=0.0503)
GRHPR | 216308_x_at
Expression level (log2)
Age
Smoker
non−smoker

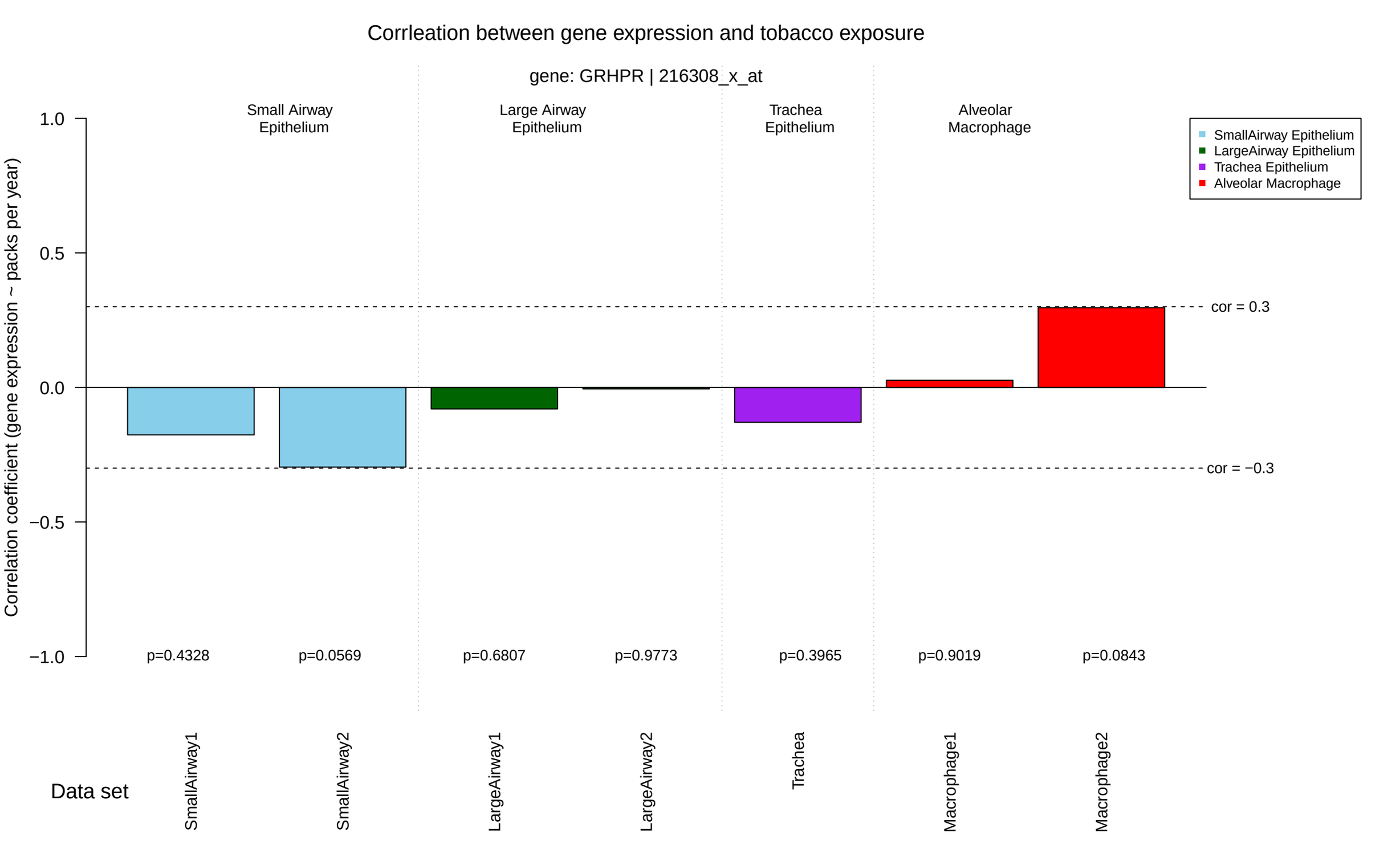

set: LargeAirway1 (p=0.0429)
GRHPR | 216308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6869)
GRHPR | 216308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3008)
GRHPR | 216308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7871)
GRHPR | 216308_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0255)
GRHPR | 216308_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRHPR | 216308_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4328
p=0.0569
p=0.6807
p=0.9773
p=0.3965
p=0.9019
p=0.0843
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2